OpenStack Labs

Lab 06: Customizing Instances

Contents

About This Document

- This document was developed by a team at the University of Tennessee at Chattanooga led by Dr. Mengjun Xie (**mengjun-xie@utc.edu**).
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Introduction

In this lab, you will use the **cloud-init** utility to customize OpenStack instances. This utility is used to perform actions from an instance upon creation. For example, it can be used to install a list of packages or change the message of the day displayed upon login to the instance.

Objectives

- Customize an instance with **cloud-init** with the *Horizon Dashboard*.
- Customize an instance with **cloud-init** with the *OpenStack Unified CLI*.
- Verify instance customization.

Lab Settings

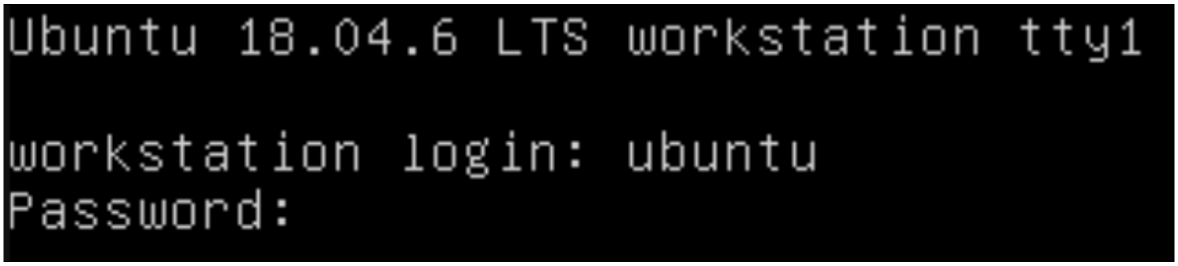
The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account	Password
workstation	ens3: 192.168.1.21 ens4: 172.25.250.21	ubuntu	ubuntu
devstack	ens3: 192.168.1.20 ens4: 172.25.250.20	ubuntu	ubuntu

1 Setting Up the Environment

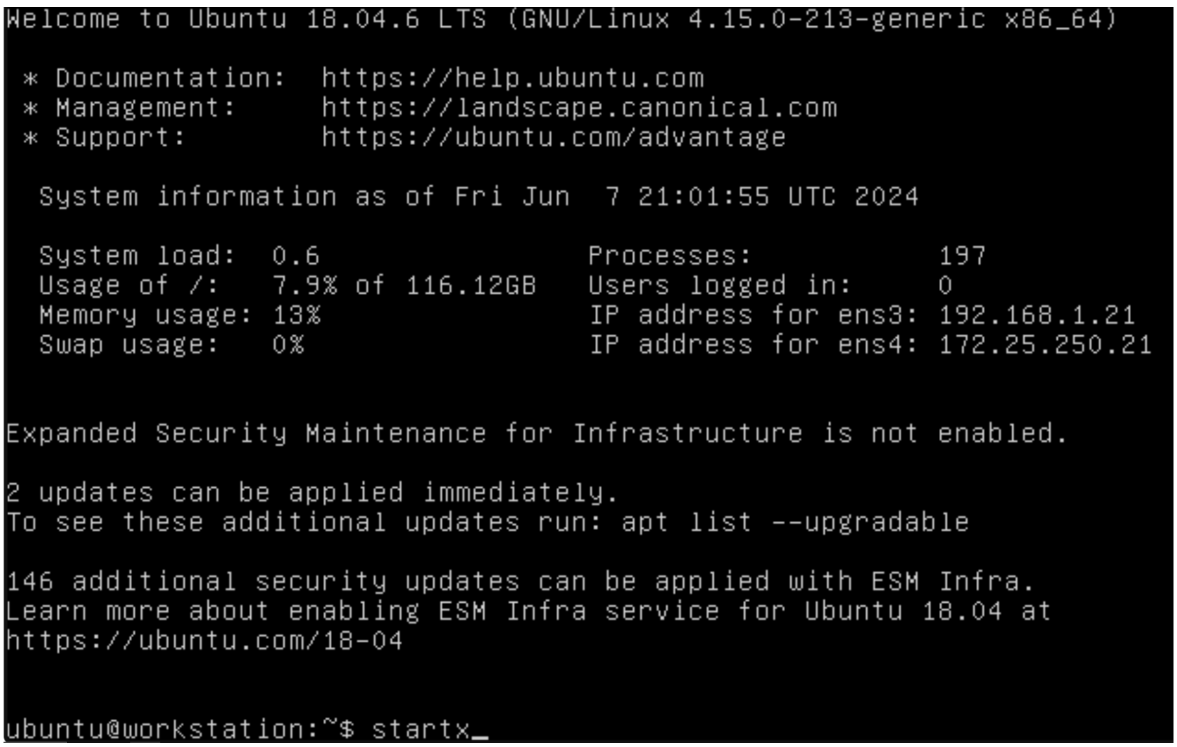
In this task, you will set up the OpenStack environment to support the customized external instances you will create in the following sections.

1.1. Log in to the **workstation** machine as the **ubuntu** user with password **ubuntu**.



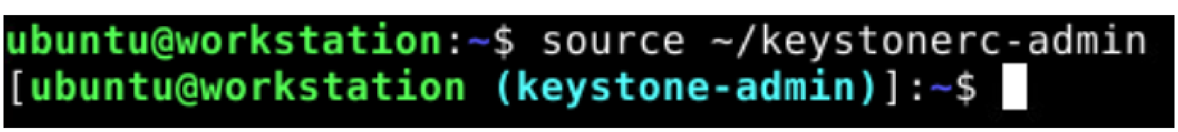
1.2. Launch the graphical user interface.

```
ubuntu@workstation:~$ startx
```



1.3. Open a terminal window and source the keystone credentials for the **admin** user.

```
ubuntu@workstation:~$ source ~/keystonerc-admin
```



1.4. In this lab, we will create our own router and external network. First, the default router and external network need to be deleted. List the routers to find the default router's name so it can be deleted.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router list
+--------------+---------+--------+-------+-------------+-------+---------------+
| ID           | Name    | Status | State | Distributed | HA    | Project       |
+--------------+---------+--------+-------+-------------+-------+---------------+
| 07df5a07-87  | router1 | ACTIVE | UP    | False       | False | 39e851b14f864 |
| 95-4d8e-     |         |        |       |             |       | 573aad60582c3 |
| accf-f9d7e4  |         |        |       |             |       | 5e40dc        |
| cbb4ee       |         |        |       |             |       |               |
+--------------+---------+--------+-------+-------------+-------+---------------+
[ubuntu@workstation (keystone-admin)]:~$
```

Tip

When typing the command, make sure there is a space between **list** and the \ character, and press **Enter** to get the > and continue typing the rest of the command.

1.5. Before the router can be deleted, it must be disconnected from all subnets. First, remove its external gateway.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router unset \
> --external-gateway \
> router1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router unset \
> --external-gateway \
> router1
[ubuntu@workstation (keystone-admin)]:~$
```

1.6. In previous labs, you disconnected routers from their subnets by using a **for** loop to iterate over interface IDs. In this lab, you will instead identify the connected subnets by name before disconnecting them. To begin, list the details of **router1** to see which subnets it is connected to.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router show router1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router show router1
+-------------------------+-----------------------------------------------------+
| Field                   | Value                                               |
+-------------------------+-----------------------------------------------------+
| admin_state_up          | UP                                                  |
| availability_zone_hints |                                                     |
| availability_zones      |                                                     |
| created_at              | 2024-02-09T19:58:10Z                                |
| description             |                                                     |
| distributed             | False                                               |
| external_gateway_info   | {"network_id":                                      |
|                         | "32da4c25-b517-40c5-97e3-cea031467d13",             |
|                         | "enable_snat": true, "external_fixed_ips":          |
|                         | [{"subnet_id": "c7916655-8954-4bf4-913d-            |
|                         | 416702f35d1b", "ip_address": "172.24.4.73"},        |
|                         | {"subnet_id": "4fc6bf88-919c-49df-                  |
|                         | 83c4-b09bd65776ad", "ip_address": "2001:db8::1"}]}  |
| flavor_id               | None                                                |
| ha                      | False                                               |
| id                      | 07df5a07-8795-4d8e-accf-f9d7e4cbb4ee                |
| interfaces_info         | [{"subnet_id": "fa8a2545-5a8c-44a2-bacc-            |
|                         | 1b86c253b880", "ip_address": "10.0.0.1",            |
|                         | "port_id":                                          |
|                         | "a05f4e1c-4014-4d25-9538-64e8114197e1"},            |
|                         | {"subnet_id": "674205b6-1357-4727-a21a-             |
|                         | 94220492a57f", "ip_address": "fd96:731b:22b0::1",   |
|                         | "port_id":                                          |
|                         | "d369b706-db4a-4239-b715-721708931870"}]            |
| name                    | router1                                             |
| project_id              | 39e851b14f864573aad60582c35e40dc                    |
| revision_number         | 6                                                   |
| routes                  |                                                     |
| status                  | ACTIVE                                              |
| tags                    |                                                     |
| updated_at              | 2024-02-09T19:58:27Z                                |
+-------------------------+-----------------------------------------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

1.7. List the subnets to see the which names are mapped to the IDs from the previous step.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack subnet list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack subnet list
+-------------------+-------------------+-------------------+---------------------+
| ID                | Name              | Network           | Subnet              |
+-------------------+-------------------+-------------------+---------------------+
| 4fc6bf88-919c-    | ipv6-public-      | 32da4c25-b517-40  | 2001:db8::/64       |
| 49df-83c4-b09bd6  | subnet            | c5-97e3-cea03146  |                     |
| 5776ad            |                   | 7d13              |                     |
| 674205b6-1357-47  | ipv6-private-     | 966ecb4f-4ff8-44  | fd96:731b:22b0::/64 |
| 27-a21a-          | subnet            | ea-a476-2d2f1895  |                     |
| 94220492a57f      |                   | 5085              |                     |
| 7e456257-76e5     | shared-subnet     | 9f23266f-d833-43  | 192.168.233.0/24    |
| -4cfc-bf3f-       |                   | 37-9a27-4818a6d2  |                     |
| b2a3876dba40      |                   | 8e9e              |                     |
| c7916655-8954-4b  | public-subnet     | 32da4c25-b517-40  | 172.24.4.0/24       |
| f4-913d-          |                   | c5-97e3-cea03146  |                     |
| 416702f35d1b      |                   | 7d13              |                     |
| fa8a2545-5a8c-    | private-subnet    | 966ecb4f-4ff8-44  | 10.0.0.0/26         |
| 44a2-bacc-        |                   | ea-a476-2d2f1895  |                     |
| 1b86c253b880      |                   | 5085              |                     |
+-------------------+-------------------+-------------------+---------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

1.8. By matching the names and IDs, you can see that the router is connected to the **private-subnet** and **ipv6-private-subnet** subnets. Remove **private-subnet** subnet from **router1**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router remove subnet \
> router1 \
> private-subnet
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router remove subnet \
> router1 \
> private-subnet
[ubuntu@workstation (keystone-admin)]:~$
```

1.9. Now, remove **ipv6-private-subnet** from **router1**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router remove subnet \
> router1 \
> ipv6-private-subnet
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router remove subnet \
> router1 \
> ipv6-private-subnet
[ubuntu@workstation (keystone-admin)]:~$
```

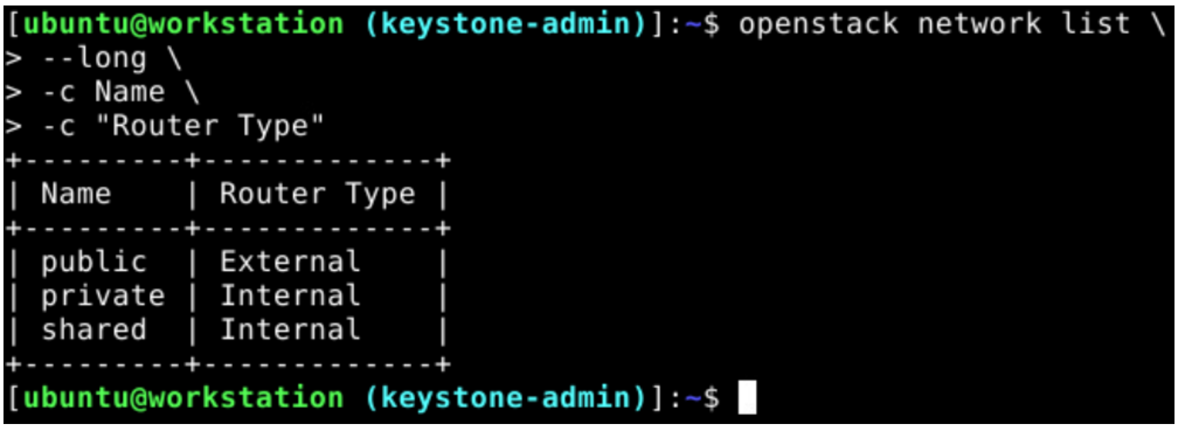
1.10. Now, the router can be deleted. Delete **router1**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router delete router1
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router delete router1
[ubuntu@workstation (keystone-admin)]:~$
```

1.11. List the details of the available networks and find which one is external. The **–long** option adds several columns to the tabular output of the command, including the **Router Type** column, which shows whether the router is internal or external. We will limit the output to just the **Name** and **Router Type** columns.

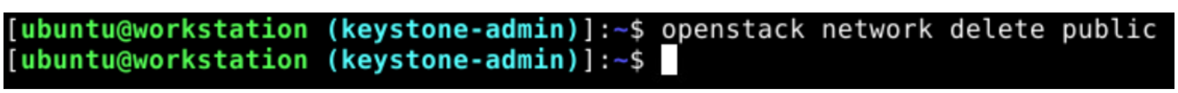
```
[ubuntu@workstation (keystone-admin)]:~$ openstack network list \
> --long \
> -c Name \
> -c "Router Type"
```



> **Tip**
>
> In most terminal commands, arguments that contain spaces must be enclosed in quotation marks so that the shell treats them as a single value. In this case, **"Router Type"** must be quoted to be interpreted correctly.

1.12. The output of the previous step shows that **public** is the external network. Delete the **public** network.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack network delete public
```



1.13. Now, we will create our own resources. First, create an external network named **extern-net**. Set the network type to **flat** and the physical network to **public**. Set the network as shared and external.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack network create \
> --external \
> --share \
> --provider-network-type flat \
> --provider-physical-network public \
> extern-net
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack network create \
> --external \
> --share \
> --provider-network-type flat \
> --provider-physical-network public \
> extern-net
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2025-07-06T20:17:03Z                 |
| description               |                                      |
| dns_domain                | None                                 |
| id                        | fcffb4d7-ee3e-4bd5-bb7b-8856f0e16ed2 |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | False                                |
| is_vlan_transparent       | None                                 |
| mtu                       | 1500                                 |
| name                      | extern-net                           |
| port_security_enabled     | True                                 |
| project_id                | 39e851b14f864573aad60582c35e40dc     |
| provider:network_type     | flat                                 |
| provider:physical_network | public                               |
| provider:segmentation_id  | None                                 |
| qos_policy_id             | None                                 |
| revision_number           | 1                                    |
| router:external           | External                             |
| segments                  | None                                 |
| shared                    | True                                 |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      |                                      |
| updated_at                | 2025-07-06T20:17:03Z                 |
+---------------------------+--------------------------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

1.14. Create a subnet named **extern-subnet** in the **extern-net** network. Give the subnet a range of **172.25.250.60** to **172.25.250.80**. Disable DHCP services for the subnet and use the address **172.25.250.254** as the gateway as well as the DNS name server.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack subnet create \
> --subnet-range 172.25.250.0/24 \
> --no-dhcp \
> --gateway 172.25.250.254 \
> --dns-nameserver 172.25.250.254 \
> --allocation-pool start=172.25.250.60,end=172.25.250.80 \
> --network extern-net \
> extern-subnet
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack subnet create \
> --subnet-range 172.25.250.0/24 \
> --no-dhcp \
> --gateway 172.25.250.254 \
> --dns-nameserver 172.25.250.254 \
> --allocation-pool start=172.25.250.60,end=172.25.250.80 \
> --network extern-net \
> extern-subnet
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| allocation_pools  | 172.25.250.60-172.25.250.80          |
| cidr              | 172.25.250.0/24                      |
| created_at        | 2025-07-06T20:18:44Z                 |
| description       |                                      |
| dns_nameservers   | 172.25.250.254                       |
| enable_dhcp       | False                                |
| gateway_ip        | 172.25.250.254                       |
| host_routes       |                                      |
| id                | fe15d8ef-d309-48fe-9b0a-e2cda8156983 |
| ip_version        | 4                                    |
| ipv6_address_mode | None                                 |
| ipv6_ra_mode      | None                                 |
| name              | extern-subnet                        |
| network_id        | fcffb4d7-ee3e-4bd5-bb7b-8856f0e16ed2 |
| project_id        | 39e851b14f864573aad60582c35e40dc     |
| revision_number   | 0                                    |
| segment_id        | None                                 |
| service_types     |                                      |
| subnetpool_id     | None                                 |
| tags              |                                      |
| updated_at        | 2025-07-06T20:18:44Z                 |
+-------------------+--------------------------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

1.15. List the floating IPs to make sure none are already allocated. The list should be empty.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip list

[ubuntu@workstation (keystone-admin)]:~$ 
```

1.16. From the floating IP pool in the **extern-net** network, create a floating IP.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip create extern-net
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip create extern-net
+---------------------+--------------------------------------+
| Field               | Value                                |
+---------------------+--------------------------------------+
| created_at          | 2025-07-07T16:31:25Z                 |
| description         |                                      |
| fixed_ip_address    | None                                 |
| floating_ip_address | 172.25.250.76                        |
| floating_network_id | 84b47575-e146-4057-b793-c9fd4625540f |
| id                  | 3b60ab3d-68a5-408f-8950-642aacf93b71 |
| name                | 172.25.250.76                        |
| port_id             | None                                 |
| project_id          | 39e851b14f864573aad60582c35e40dc     |
| qos_policy_id       | None                                 |
| revision_number     | 0                                    |
| router_id           | None                                 |
| status              | DOWN                                 |
| subnet_id           | None                                 |
| updated_at          | 2025-07-07T16:31:25Z                 |
+---------------------+--------------------------------------+
[ubuntu@workstation (keystone-admin)]:~$ 
```

1.17. Create a router named **extern-router**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router create extern-router
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router create extern-router
+-------------------------+--------------------------------------+
| Field                   | Value                                |
+-------------------------+--------------------------------------+
| admin_state_up          | UP                                   |
| availability_zone_hints |                                      |
| availability_zones      |                                      |
| created_at              | 2024-06-20T00:40:08Z                 |
| description             |                                      |
| distributed             | False                                |
| external_gateway_info   | None                                 |
| flavor_id               | None                                 |
| ha                      | False                                |
| id                      | c92e57ef-ba53-4081-92be-d4856d09ae1b |
| name                    | extern-router                        |
| project_id              | 39e851b14f864573aad60582c35e40dc     |
| revision_number         | 1                                    |
| routes                  |                                      |
| status                  | ACTIVE                               |
| tags                    |                                      |
| updated_at              | 2024-06-20T00:40:08Z                 |
+-------------------------+--------------------------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

1.18. List the available subnets to see what the router can be connected to.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack subnet list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack subnet list
+-------------------+-------------------+-------------------+-------------------+
| ID                | Name              | Network           | Subnet            |
+-------------------+-------------------+-------------------+-------------------+
| 674205b6-1357-472 | ipv6-private-     | 966ecb4f-4ff8-44e | fd96:731b:22b0::/ |
| 7-a21a-           | subnet            | a-a476-2d2f189550 | 64                |
| 94220492a57f      |                   | 85                |                   |
| 7e456257-76e5     | shared-subnet     | 9f23266f-d833-433 | 192.168.233.0/24  |
| -4cfc-bf3f-       |                   | 7-9a27-4818a6d28e |                   |
| b2a3876dba40      |                   | 9e                |                   |
| fa8a2545-5a8c-    | private-subnet    | 966ecb4f-4ff8-44e | 10.0.0.0/26       |
| 44a2-bacc-        |                   | a-a476-2d2f189550 |                   |
| 1b86c253b880      |                   | 85                |                   |
| fe15d8ef-d309     | extern-subnet     | fcffb4d7-ee3e-    | 172.25.250.0/24   |
| -48fe-9b0a-       |                   | 4bd5-bb7b-        |                   |
| e2cda8156983      |                   | 8856f0e16ed2      |                   |
+-------------------+-------------------+-------------------+-------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

1.19. Connect the router to the **shared-subnet** subnet.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router add subnet \
> extern-router shared-subnet
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router add subnet \
> extern-router shared-subnet
[ubuntu@workstation (keystone-admin)]:~$
```

1.20. Set the **extern-net** network as the gateway for the router.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router set \
> --external-gateway extern-net \
> extern-router
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack router set \
> --external-gateway extern-net \
> extern-router
[ubuntu@workstation (keystone-admin)]:~$
```

1.21. List the current key pairs to make sure there are not any already allocated. The list should be empty.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair list

[ubuntu@workstation (keystone-admin)]:~$
```

1.22. Create the key pair **dev-keypair** and save the private key to the file ~**/Downloads/dev-keypair.pem**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair create \
> dev-keypair > ~/Downloads/dev-keypair.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack keypair create \
> dev-keypair > ~/Downloads/dev-keypair.pem
[ubuntu@workstation (keystone-admin)]:~$
```

1.23. View the permission of the **/Downloads/dev-keypair.pem** file. You can see that the **ubuntu** user has read and write privileges to the file, the **ubuntu** user group has read and write privileges to the file, and other users have read permissions to the file. However, this is a sensitive file, and only the **ubuntu** user should have any privileges to it.

```
[ubuntu@workstation (keystone-admin)]:~$ ls -l ~/Downloads/dev-keypair.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ ls -l ~/Downloads/dev-keypair.pem
-rw-rw-r-- 1 ubuntu ubuntu 1680 Jun 20 00:50 /home/ubuntu/Downloads/dev-keypair.pem
[ubuntu@workstation (keystone-admin)]:~$
```

1.24. Use the **chmod** command with a mode of **600** to make it so that the **ubuntu** user has read/write permissions on the file, and groups and other users have no permissions to the file.

```
[ubuntu@workstation (keystone-admin)]:~$ chmod 600 ~/Downloads/dev-keypair.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ chmod 600 ~/Downloads/dev-keypair.pem
[ubuntu@workstation (keystone-admin)]:~$
```

1.25. List the permissions again to ensure that the change took effect.

```
[ubuntu@workstation (keystone-admin)]:~$ ls -l ~/Downloads/dev-keypair.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ ls -l ~/Downloads/dev-keypair.pem
-rw------- 1 ubuntu ubuntu 1680 Jun 20 00:50 /home/ubuntu/Downloads/dev-keypair.pem
[ubuntu@workstation (keystone-admin)]:~$
```

1.26. Create the **dev-secgroup** security group.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group create \
> dev-secgroup
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group create \
> dev-secgroup
+-----------------+----------------------------------------------------------------+
| Field           | Value                                                          |
+-----------------+----------------------------------------------------------------+
| created_at      | 2025-07-06T20:21:05Z                                           |
| description     | dev-secgroup                                                   |
| id              | 69c4fd3c-a78c-44b0-9c4b-03a7101d65c2                           |
| name            | dev-secgroup                                                   |
| project_id      | 39e851b14f864573aad60582c35e40dc                               |
| revision_number | 1                                                              |
| rules           | created_at='2025-07-06T20:21:05Z', direction='egress',         |
|                 | ethertype='IPv4', id='3e3d08e8-2688-4354-a20a-                 |
|                 | 44f062537880', standard_attr_id='48',                          |
|                 | updated_at='2025-07-06T20:21:05Z'                              |
|                 | created_at='2025-07-06T20:21:05Z', direction='egress',         |
|                 | ethertype='IPv6',                                              |
|                 | id='4ba2c126-18ee-4739-9d23-477ff8677af1',                     |
|                 | standard_attr_id='47', updated_at='2025-07-06T20:21:05Z'       |
| updated_at      | 2025-07-06T20:21:05Z                                           |
+-----------------+----------------------------------------------------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

1.27. List the default security group rules.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> dev-secgroup
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> dev-secgroup
+--------------------+-------------+----------+------------+-----------------------+
| ID                 | IP Protocol | IP Range | Port Range | Remote Security Group |
+--------------------+-------------+----------+------------+-----------------------+
| 3e3d08e8-2688-4354 | None        | None     |            | None                  |
| -a20a-44f062537880 |             |          |            |                       |
| 4ba2c126-18ee-4739 | None        | None     |            | None                  |
| -9d23-477ff8677af1 |             |          |            |                       |
+--------------------+-------------+----------+------------+-----------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

Tip

You can verify that the default rules allow any outgoing traffic over IPv4 or IPv6 with this command:

```
openstack security group rule show <rule-id>
```

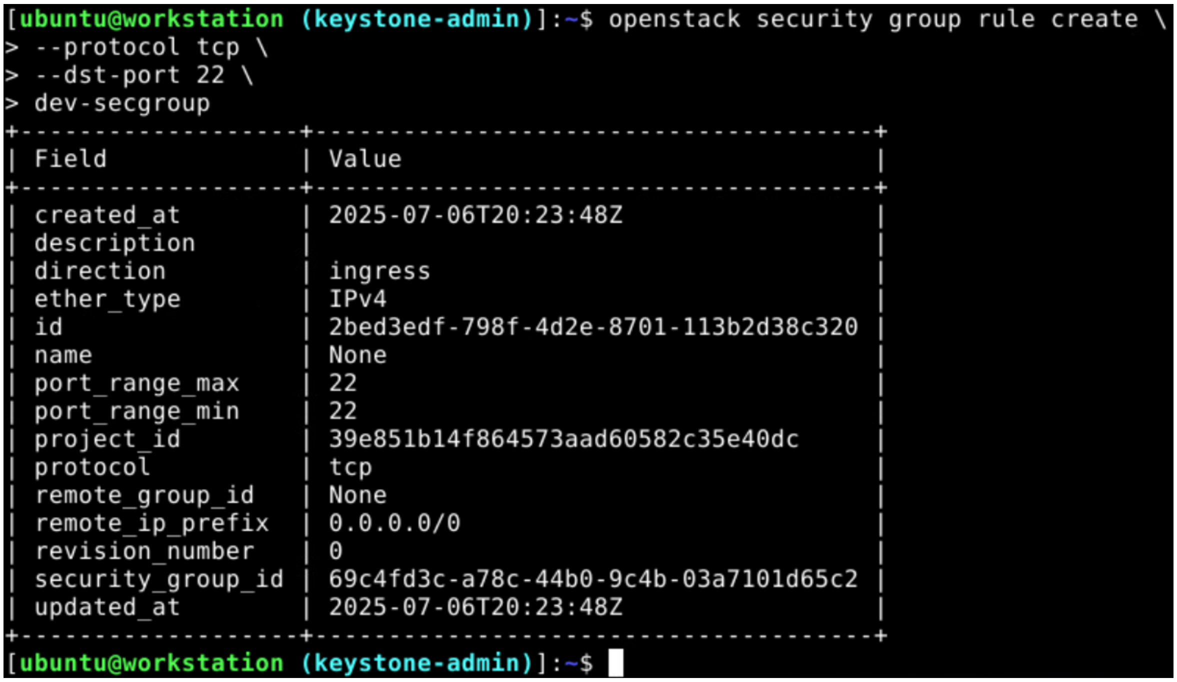
1.28. Add a security rule in the **dev-secgroup** security group to allow remote ICMP traffic.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol icmp \
> dev-secgroup
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol icmp \
> dev-secgroup
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| created_at        | 2025-07-06T20:23:12Z                 |
| description       |                                      |
| direction         | ingress                              |
| ether_type        | IPv4                                 |
| id                | b05d62d2-b94f-443c-92b8-1086bbd59b88 |
| name              | None                                 |
| port_range_max    | None                                 |
| port_range_min    | None                                 |
| project_id        | 39e851b14f864573aad60582c35e40dc     |
| protocol          | icmp                                 |
| remote_group_id   | None                                 |
| remote_ip_prefix  | 0.0.0.0/0                            |
| revision_number   | 0                                    |
| security_group_id | 69c4fd3c-a78c-44b0-9c4b-03a7101d65c2 |
| updated_at        | 2025-07-06T20:23:12Z                 |
+-------------------+--------------------------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

1.29. Add another security rule to allow remote connection using SSH on the default port 22.

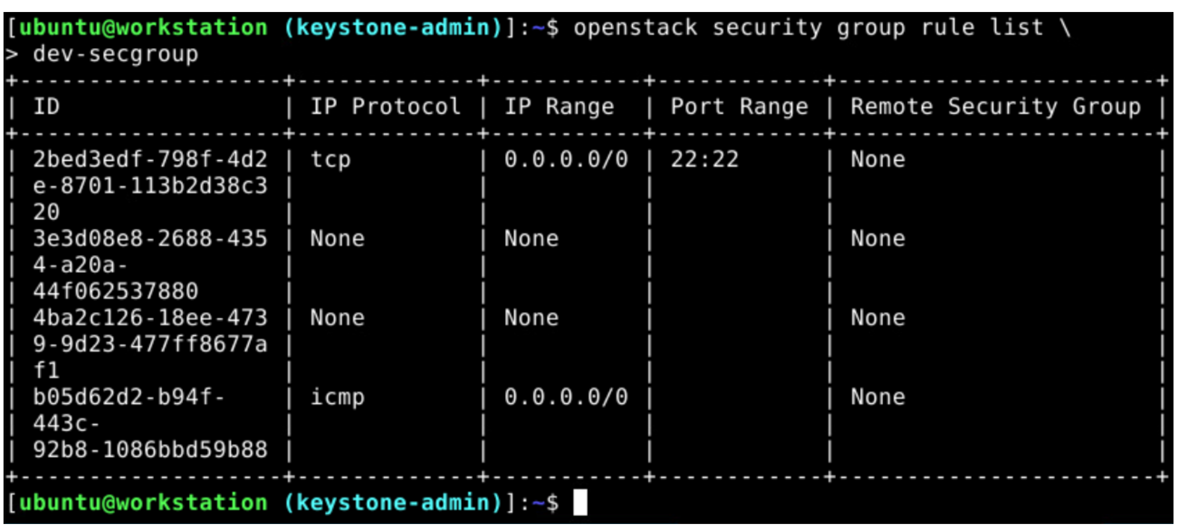
```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol tcp \
> --dst-port 22 \
> dev-secgroup
```



```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule create \
> --protocol tcp \
> --dst-port 22 \
> dev-secgroup
+-------------------+--------------------------------------+
| Field             | Value                                |
+-------------------+--------------------------------------+
| created_at        | 2025-07-06T20:23:48Z                 |
| description       |                                      |
| direction         | ingress                              |
| ether_type        | IPv4                                 |
| id                | 2bed3edf-798f-4d2e-8701-113b2d38c320 |
| name              | None                                 |
| port_range_max    | 22                                   |
| port_range_min    | 22                                   |
| project_id        | 39e851b14f864573aad60582c35e40dc     |
| protocol          | tcp                                  |
| remote_group_id   | None                                 |
| remote_ip_prefix  | 0.0.0.0/0                            |
| revision_number   | 0                                    |
| security_group_id | 69c4fd3c-a78c-44b0-9c4b-03a7101d65c2 |
| updated_at        | 2025-07-06T20:23:48Z                 |
+-------------------+--------------------------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

1.30. List the security group rules again to see that the new rules have taken effect.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> dev-secgroup
```



```
[ubuntu@workstation (keystone-admin)]:~$ openstack security group rule list \
> dev-secgroup
+-------------------+-------------+-----------+------------+-----------------------+
| ID                | IP Protocol | IP Range  | Port Range | Remote Security Group |
+-------------------+-------------+-----------+------------+-----------------------+
| 2bed3edf-798f-4d2 | tcp         | 0.0.0.0/0 | 22:22      | None                  |
| e-8701-113b2d38c3 |             |           |            |                       |
| 20                |             |           |            |                       |
| 3e3d08e8-2688-435 | None        | None      |            | None                  |
| 4-a20a-           |             |           |            |                       |
| 44f062537880      |             |           |            |                       |
| 4ba2c126-18ee-473 | None        | None      |            | None                  |
| 9-9d23-477ff8677a |             |           |            |                       |
| f1                |             |           |            |                       |
| b05d62d2-b94f-    | icmp        | 0.0.0.0/0 |            | None                  |
| 443c-             |             |           |            |                       |
| 92b8-1086bbd59b88 |             |           |            |                       |
+-------------------+-------------+-----------+------------+-----------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

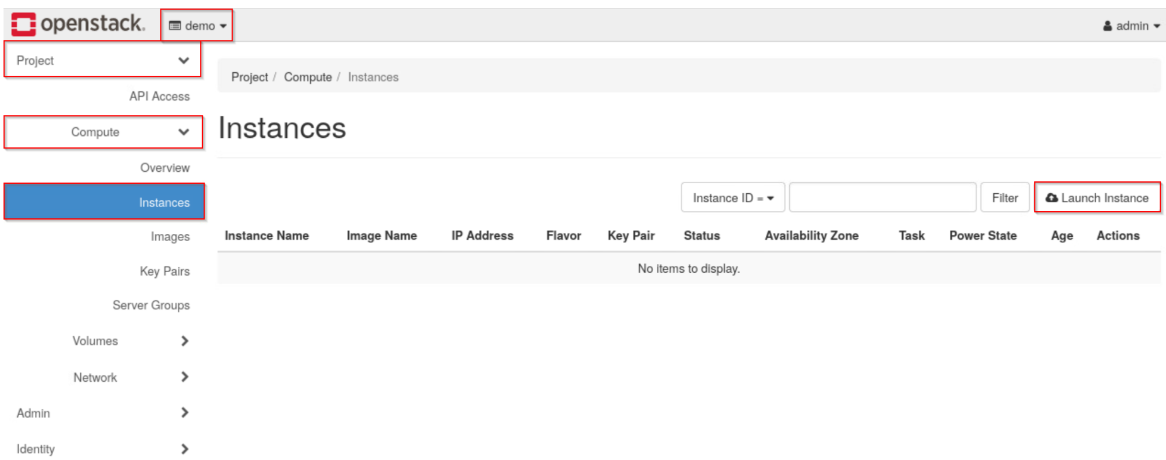
1.31. Close the terminal window, and continue to the next task.

2 Creating a Customized Instance with the Horizon Dashboard

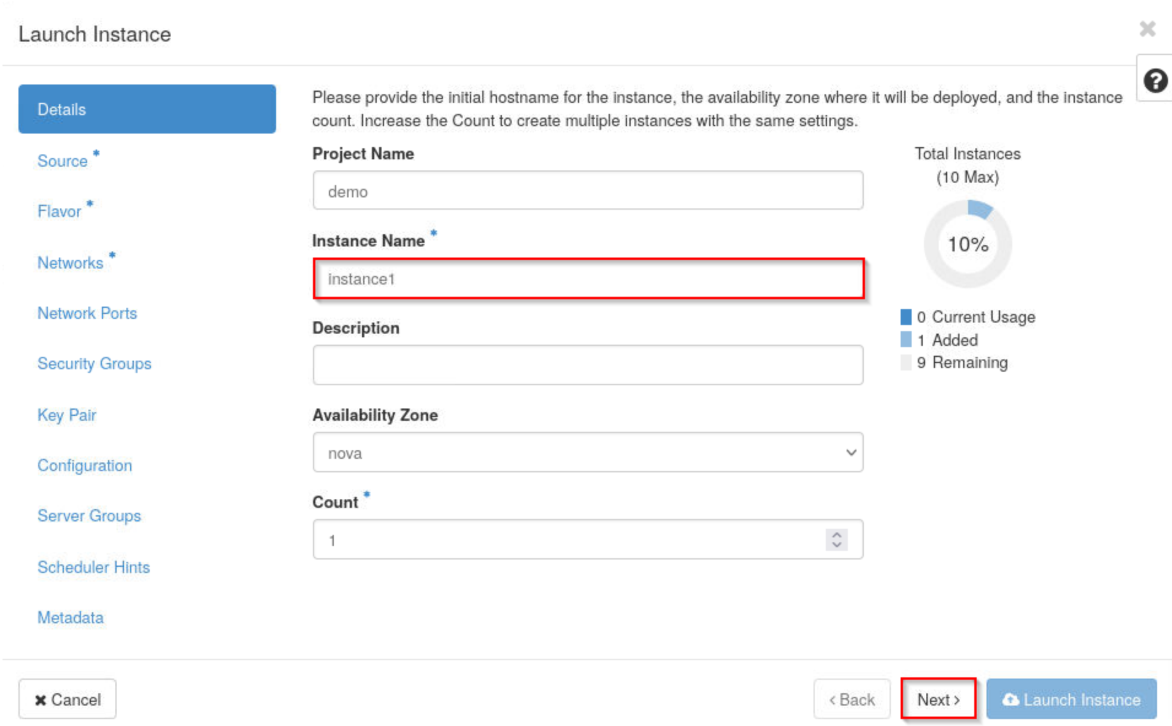
In this task, you will create a customized instance with the Horizon Dashboard and log in to the instance to verify that the customization took effect.

2.1. Open the web browser, and navigate to **192.168.1.20**. Log in to the dashboard as **admin** with the password **secret**.

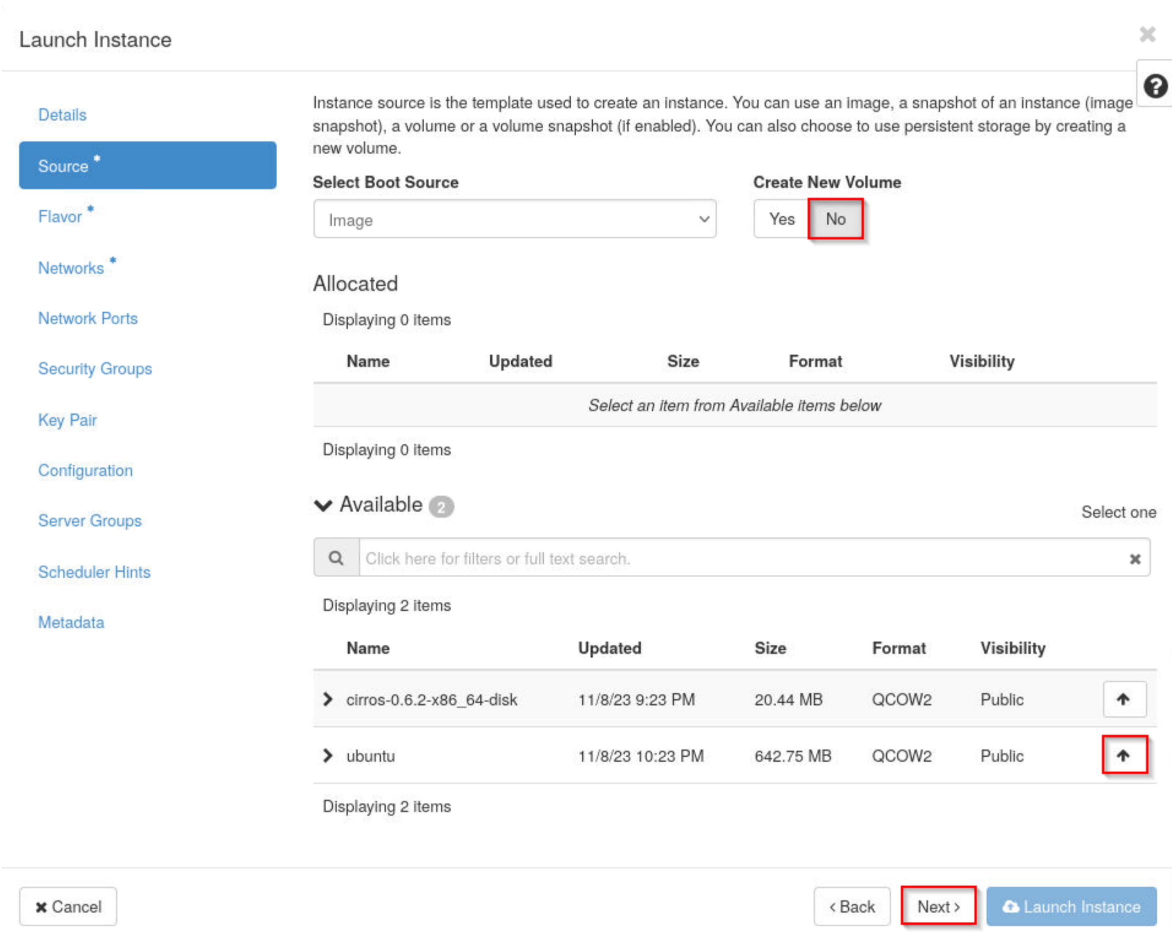
2.2. Make sure the **demo** project is selected. Navigate to **Project** > **Compute** > **Instances**, and click **Launch Instance**.



2.3. In the *Details* tab, enter **instance1** in the *Instance Name* field and click **Next**.



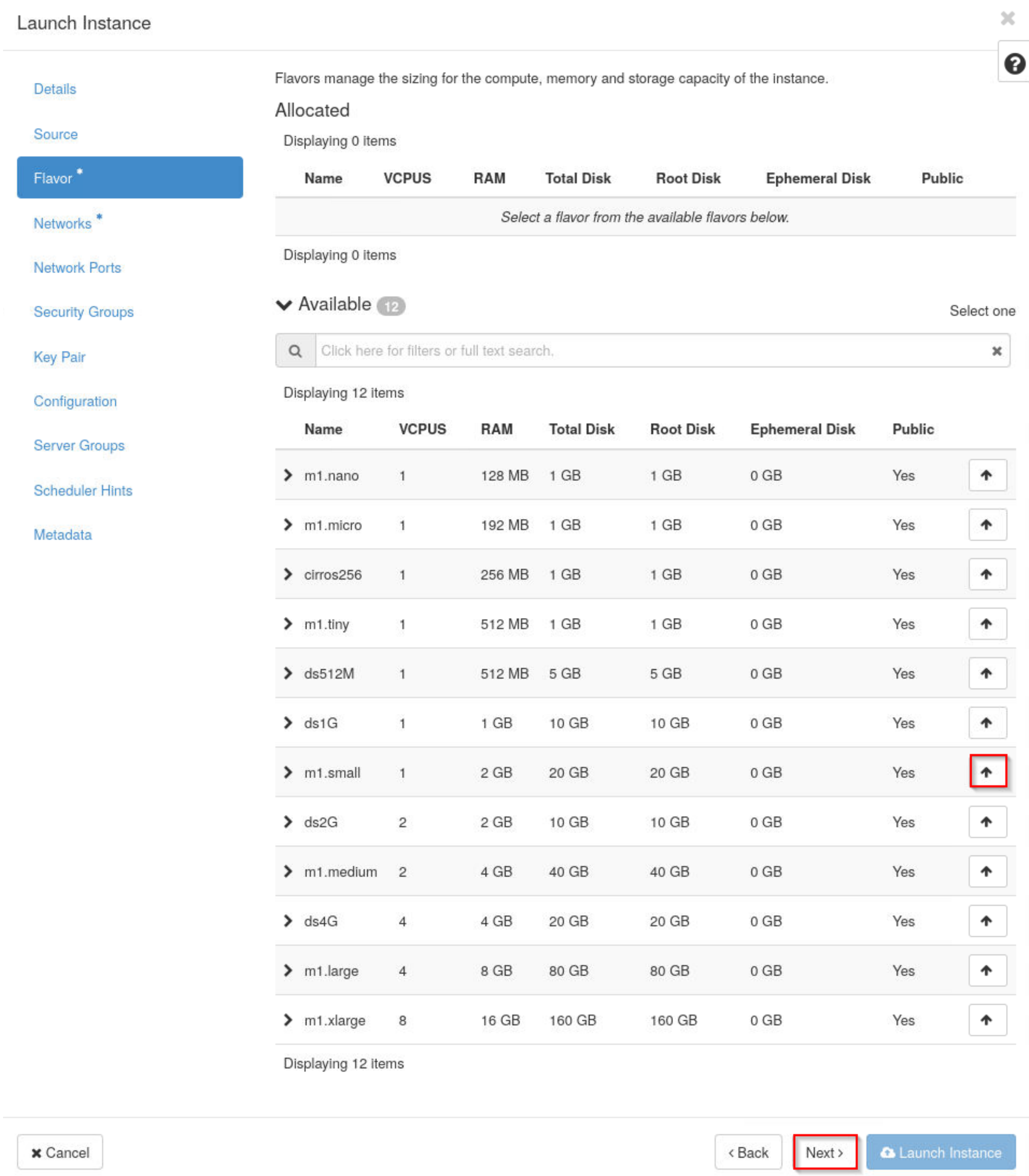
2.4. In the *Source* tab, make sure **Image** is selected in the *Select Boot Source* dropdown and click **No** under *Create New Volume*. Select the **ubuntu** image by clicking the ↑ symbol in the same row. Click **Next**.



Stop

Before proceeding to the next step, confirm that **ubuntu** appears underneath the *Allocated* section.

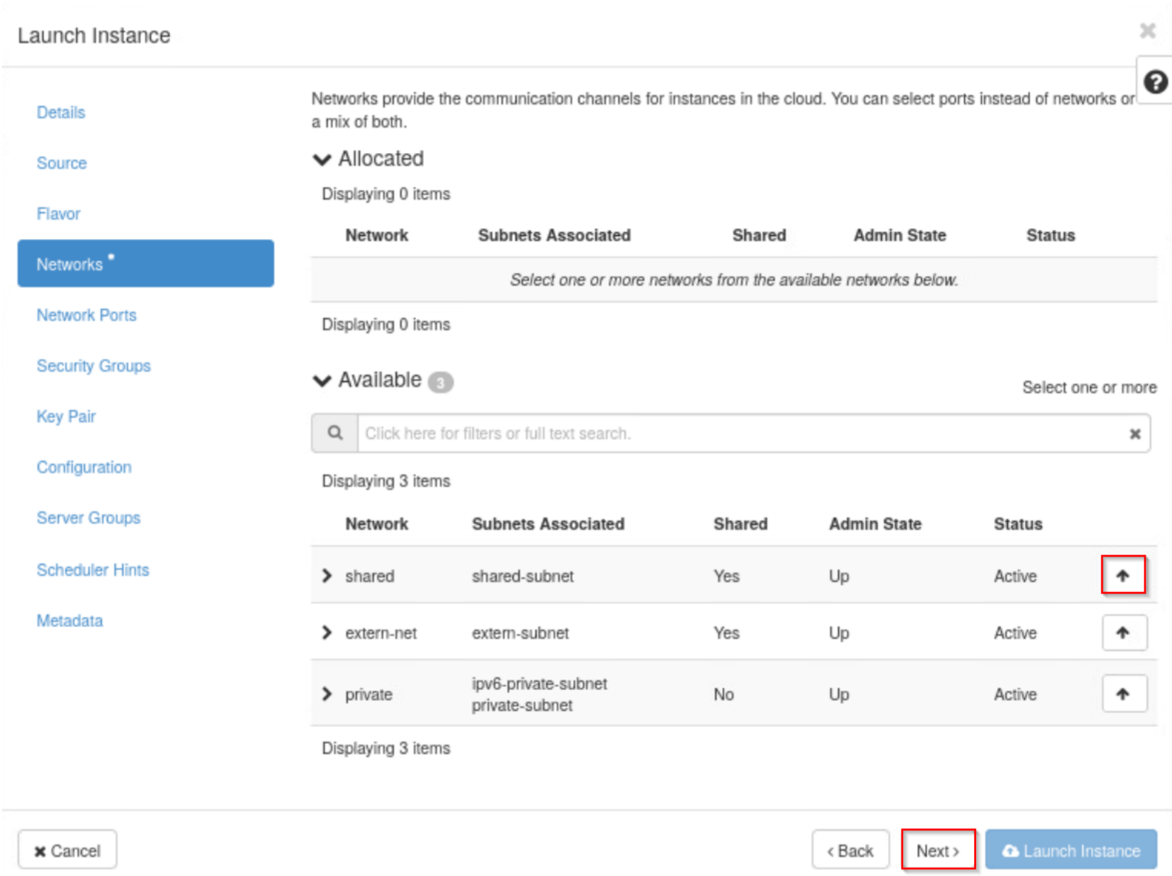
2.5. In the *Flavor* tab, click the ↑ symbol in the same row as **m1.small** Click **Next**.



Stop

Before proceeding to the next step, confirm that **m1.small** appears underneath the *Allocated* section.

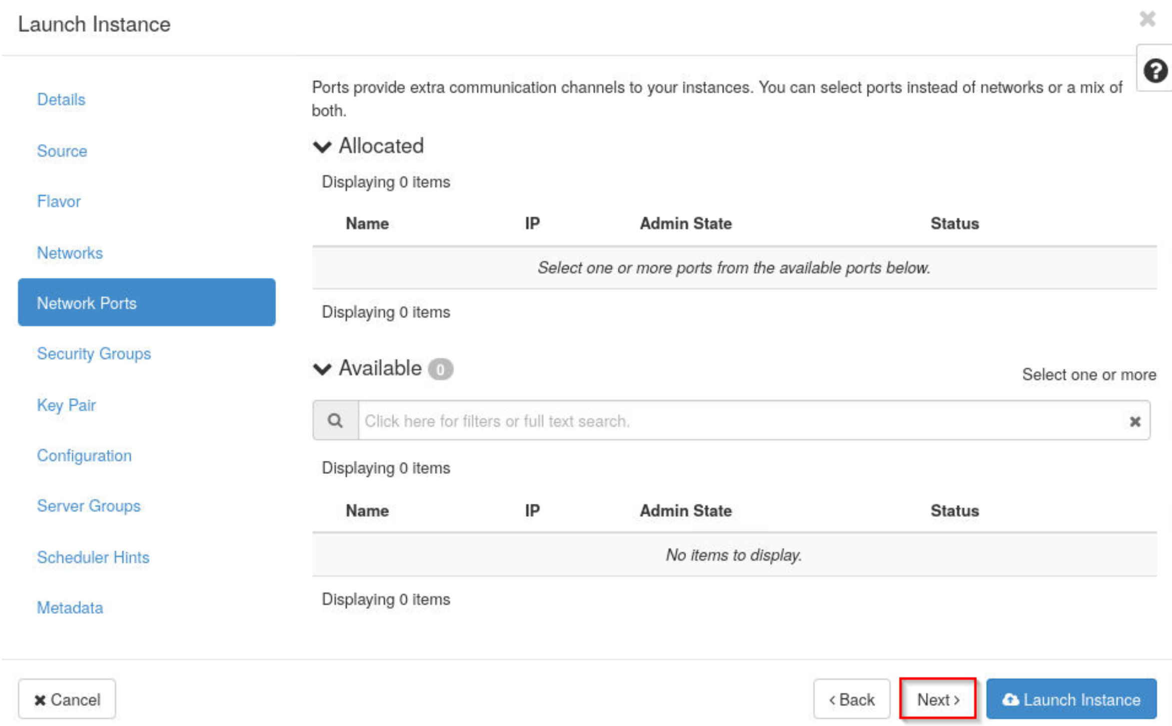
2.6. In the *Networks* tab, click the ↑ symbol in the same row as **shared**. Click **Next**.



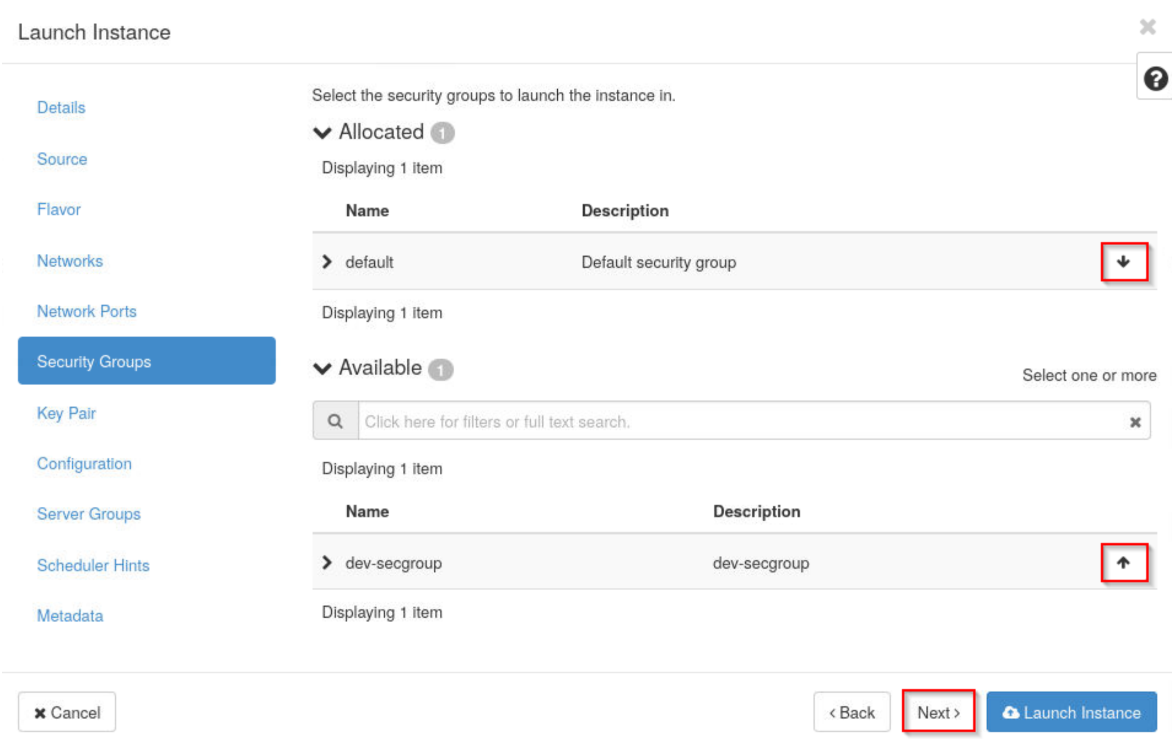
Stop

Before proceeding to the next step, confirm that **shared** appears underneath the *Allocated* section.

2.7. In the *Network Ports* tab, click **Next**.



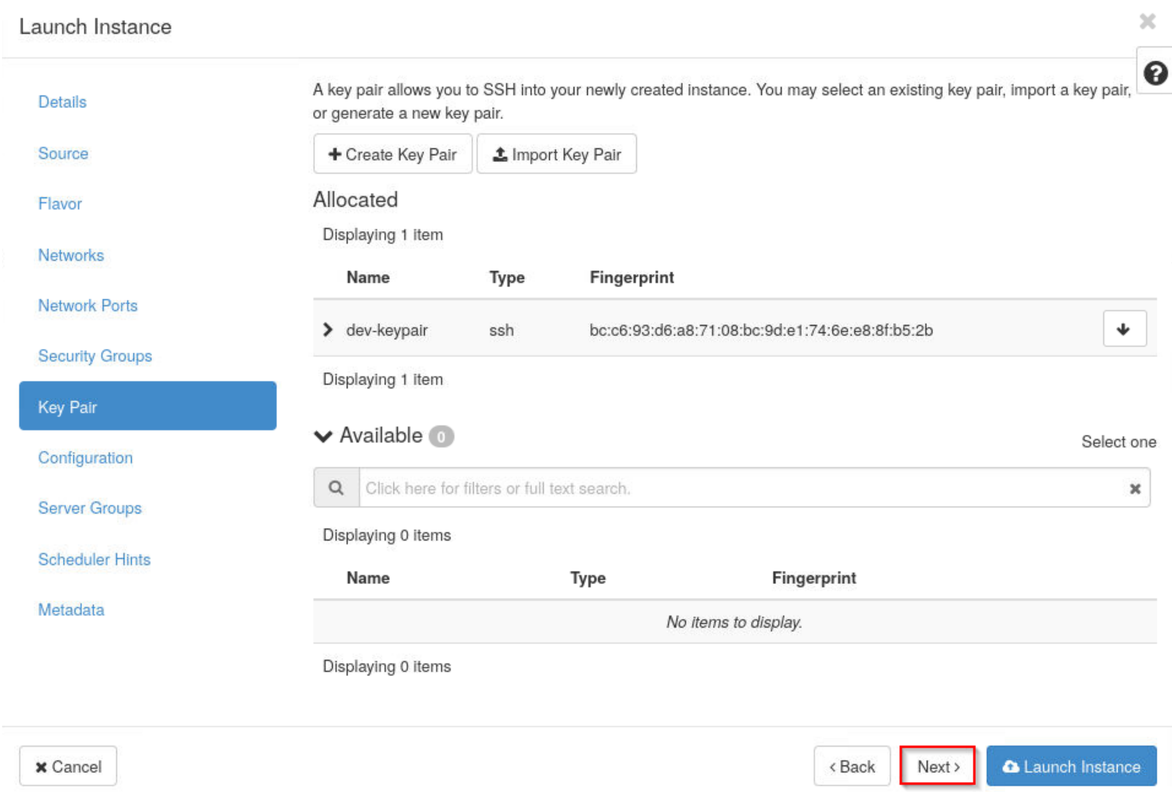
2.8. In the *Security Groups* tab, click the ↓ symbol in the same row as **default**, and click the ↑ symbol in the same row as **dev-secgroup**. Click **Next**.



Stop

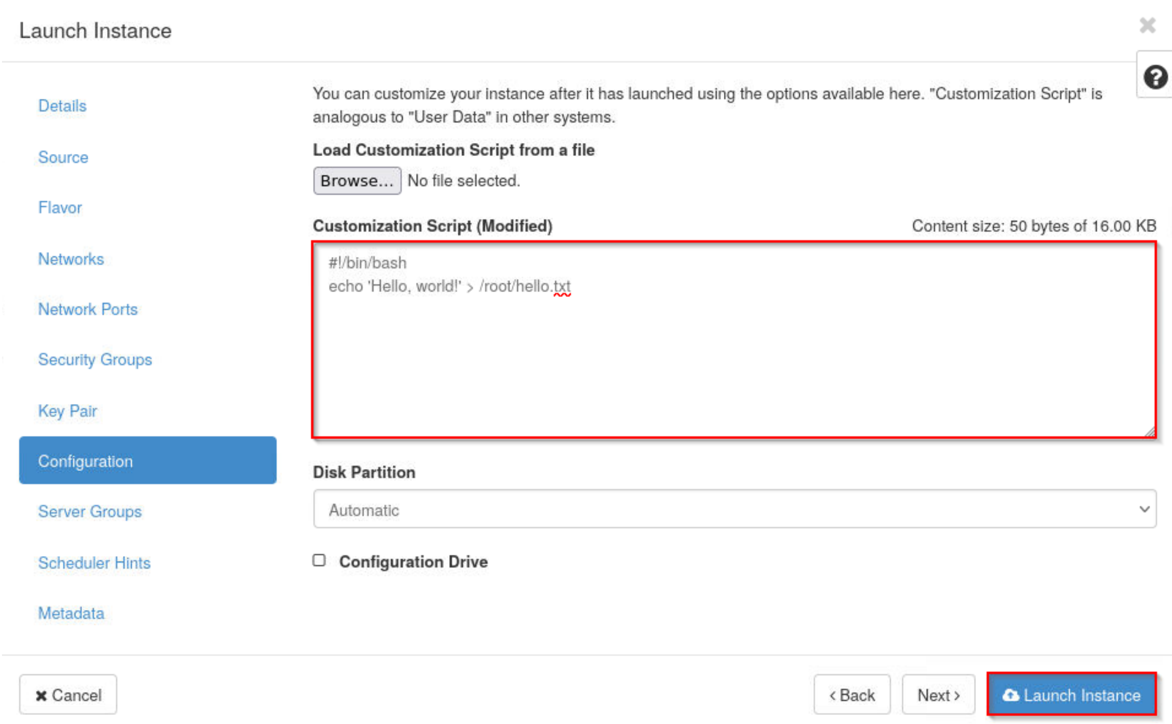
Before proceeding to the next step, confirm that only **dev-secgroup** appears underneath the *Allocated* section.

2.9. In the *Key Pair* tab, ensure that the key pair **dev-keypair** has been selected and is underneath the *Allocated* section. Click **Next**.



2.10. In the *Configuration* tab, populate the **Customization Script** field with the content below. Once finished, click **Launch Instance**.

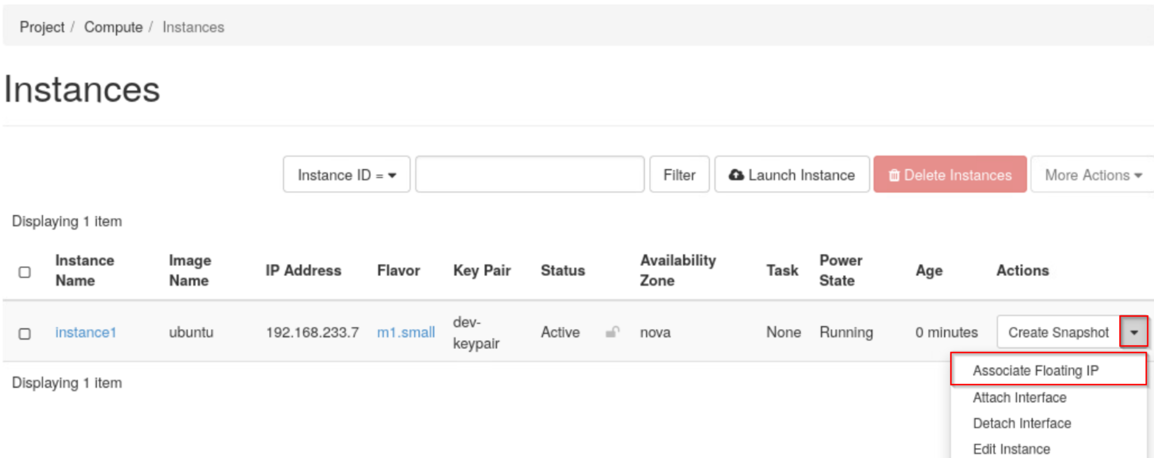
```
#!/bin/bash
echo 'Hello, world!' > /root/hello.txt
```



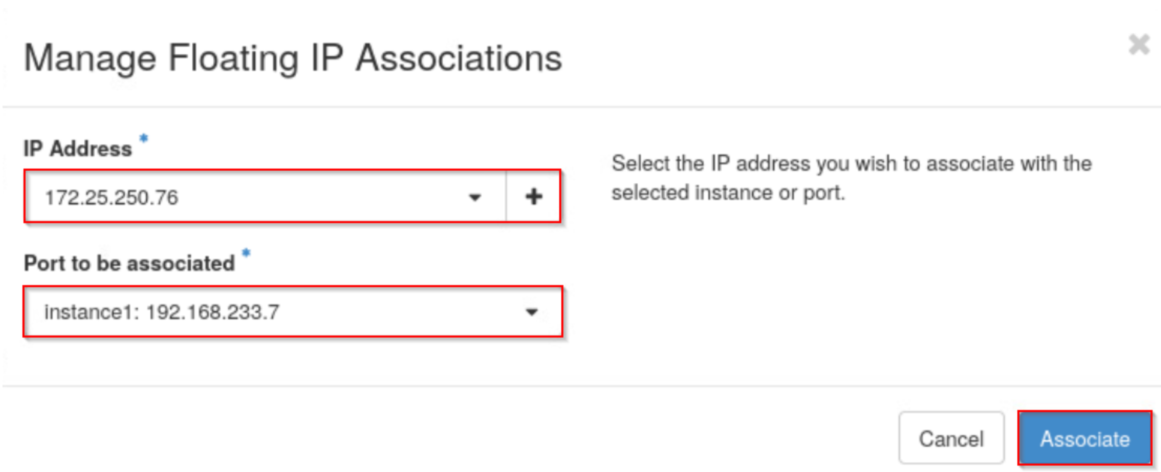
Tip

A customization script can be used to perform many commands automatically upon instance creation, such as installing packages, configuring a host name, etc. The simple script above is just an example.

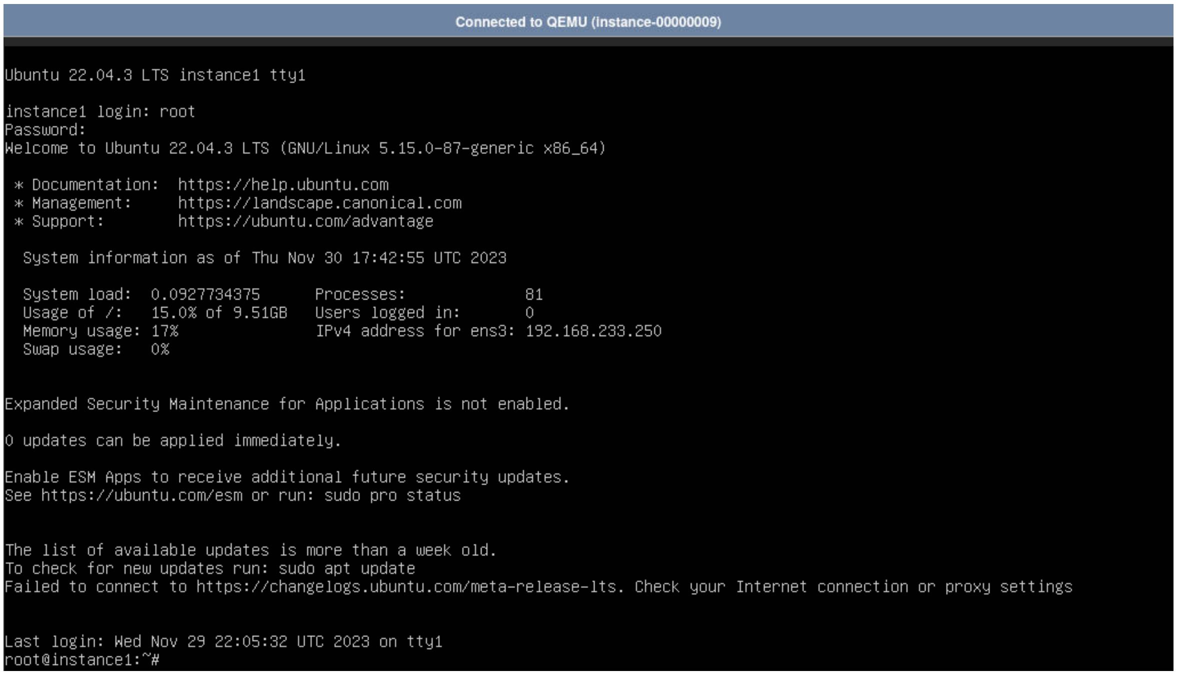
2.11. Once the status for **instance1** is **Active**, attach a floating IP address to it. Select **Associate Floating IP** from the dropdown menu next to **Create Snapshot** in the row for the instance.



2.12. Select any one of the IP addresses from the *IP Address* dropdown and select **instance1: 192.168.233.XYZ** as the *Port to be associated*. Click **Associate**.



2.13. To verify that the customization script worked, first click **instance1** under the *Instance Name* column, then navigate to the *Console* tab if you are not directed there automatically. Click the **Click here to show only the console** link. Log in to the instance as **root** with the password **secret**.



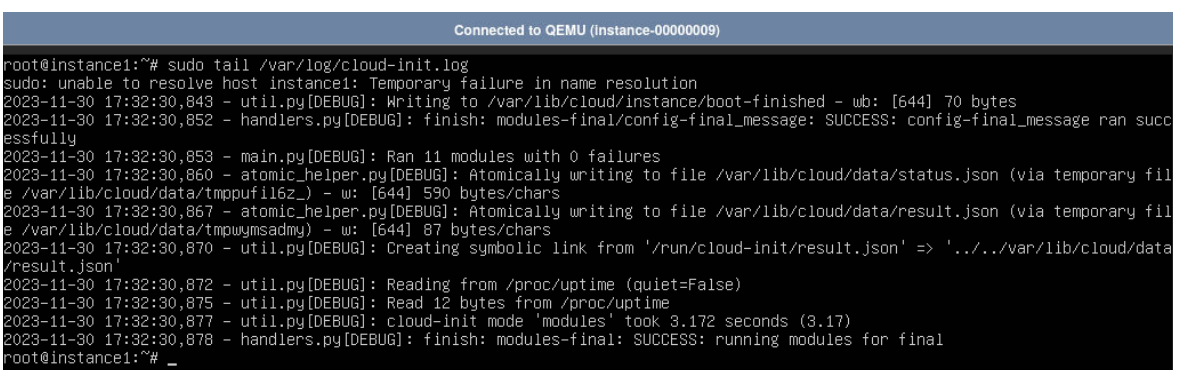
```
Connected to QEMU (instance-00000009)

Ubuntu 22.04.3 LTS instance1 tty1

instance1 login: root
Password:
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Thu Nov 30 17:42:55 UTC 2023

  System load:  0.0927734375      Processes:             81
  Usage of /:   15.0% of 9.51GB   Users logged in:       0
  Memory usage: 17%               IPv4 address for ens3: 192.168.233.250
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings


Last login: Wed Nov 29 22:05:32 UTC 2023 on tty1
root@instance1:~#
```

Note

Be patient for the login prompt. Even if OpenStack reports the instance to be active, it may still take several minutes to fully launch and present the login prompt.

2.14. Check **/var/log/cloud-init.log** to confirm that **cloud-init** ran. Use the **tail** command to print the last 10 lines of the log.

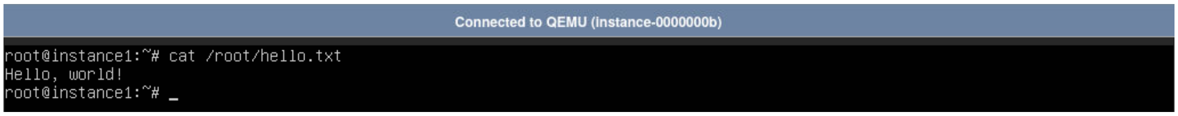
```
root@instance1:~# sudo tail /var/log/cloud-init.log
```



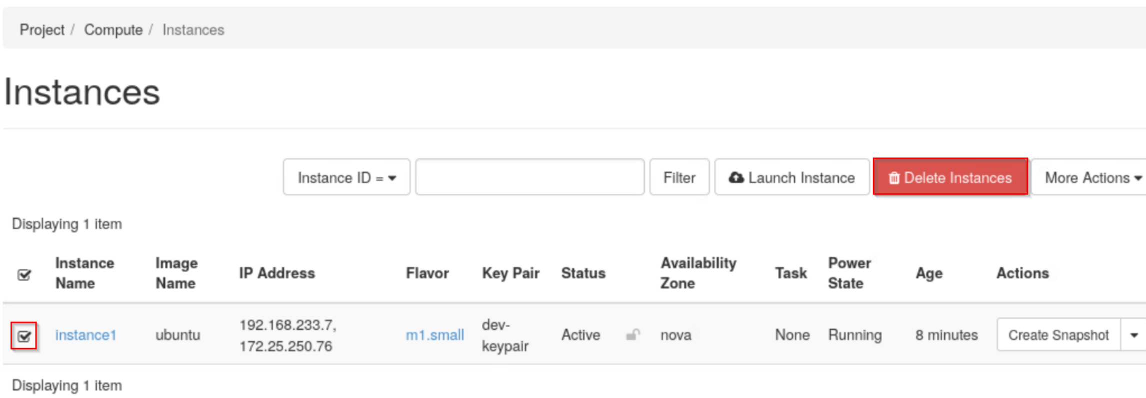
```
Connected to QEMU (instance-00000009)
root@instance1:~# sudo tail /var/log/cloud-init.log
sudo: unable to resolve host instance1: Temporary failure in name resolution
2023-11-30 17:32:30,843 - util.py[DEBUG]: Writing to /var/lib/cloud/instance/boot-finished - wb: [644] 70 bytes
2023-11-30 17:32:30,852 - handlers.py[DEBUG]: finish: modules-final/config-final_message: SUCCESS: config-final_message ran succ
essfully
2023-11-30 17:32:30,853 - main.py[DEBUG]: Ran 11 modules with 0 failures
2023-11-30 17:32:30,860 - atomic_helper.py[DEBUG]: Atomically writing to file /var/lib/cloud/data/status.json (via temporary fil
e /var/lib/cloud/data/tmppufil6z_) - w: [644] 590 bytes/chars
2023-11-30 17:32:30,867 - atomic_helper.py[DEBUG]: Atomically writing to file /var/lib/cloud/data/result.json (via temporary fil
e /var/lib/cloud/data/tmpwymsadmy) - w: [644] 87 bytes/chars
2023-11-30 17:32:30,870 - util.py[DEBUG]: Creating symbolic link from '/run/cloud-init/result.json' => '../../var/lib/cloud/data
/result.json'
2023-11-30 17:32:30,872 - util.py[DEBUG]: Reading from /proc/uptime (quiet=False)
2023-11-30 17:32:30,875 - util.py[DEBUG]: Read 12 bytes from /proc/uptime
2023-11-30 17:32:30,877 - util.py[DEBUG]: cloud-init mode 'modules' took 3.172 seconds (3.17)
2023-11-30 17:32:30,878 - handlers.py[DEBUG]: finish: modules-final: SUCCESS: running modules for final
root@instance1:~# _
```

2.15. Ensure that the **/root/hello.txt** file exists and has the correct content.

```
root@instance1:~# cat /root/hello.txt
```



2.16. Close the tab with the terminal window, and navigate back to **Project** > **Compute** > **Instances** if you are not already on that page. We are finished with this instance and will create another one in the next section. Select the checkbox next to **instance1** and click **Delete Instances**.



2.17. Log out of the *Horizon Dashboard* and close the web browser.

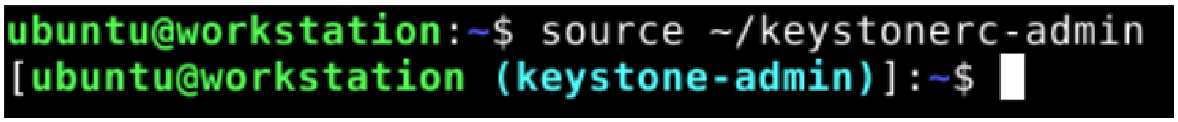
2.18. Continue to the next task.

3 Creating a Customized Instance with the OpenStack Unified CLI

In this task, you will verify that **cloud-init** has correctly customized the two instances created in the previous section.

3.1. If a terminal window is not already open, open one and source the admin credentials from the ~**/keystonerc-admin** file.

```
[ubuntu@workstation (keystone-admin)]:~$ source ~/keystonerc-admin
```



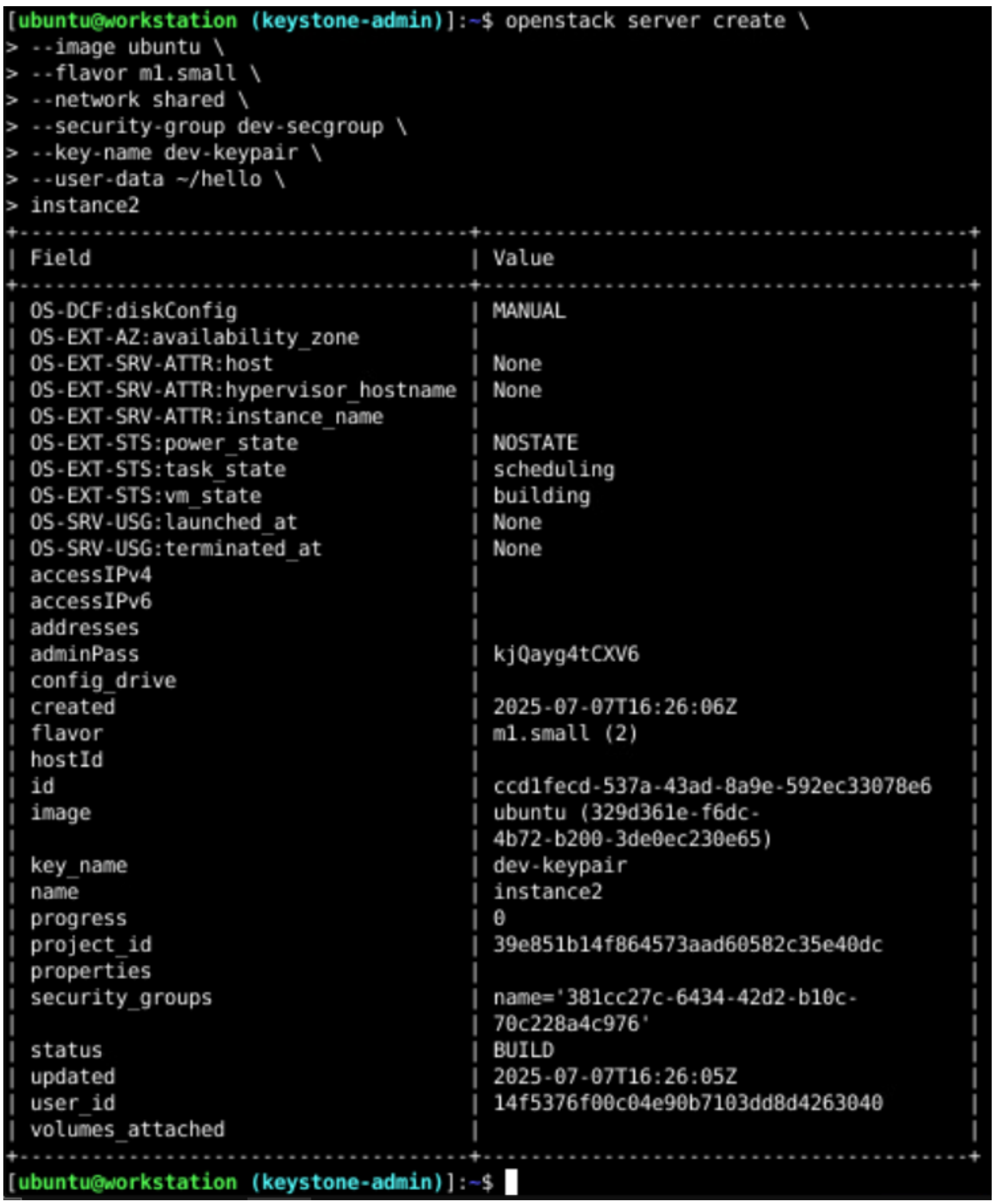
3.2. Another instance will be created and customized with the *OpenStack Unified CLI*. First, create a **user-data** script that will be attached to the instance at creation. Create a script called ~**/hello** that matches the content shown below. When the instance runs this script, it will create the file **/root/hello.txt** with the contents **Hello, world!**, and it will append the line **127.0.1.1 instance2** to the **/etc/hosts** file. This simply suppresses a warning when issuing a command with root privileges (i.e., **sudo ...**). Press **CTRL+X**, then **Y** to accept the file changes. Press **Enter** to confirm and exit back to the terminal.

```
[ubuntu@workstation (keystone-admin)]:~$ nano ~/hello
```

```
#!/bin/bash
echo 'Hello, world!' > /root/hello.txt
echo '127.0.1.1 instance2' >> /etc/hosts
```

3.3. Launch an instance with the **user-data** option with the previously created script to perform the customization. Use the **ubuntu** image, the **m1.small** flavor, the **shared** network, the **dev-secgroup** security group, and the **dev-keypair** key pair.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server create \
> --image ubuntu \
> --flavor m1.small \
> --network shared \
> --security-group dev-secgroup \
> --key-name dev-keypair \
> --user-data ~/hello \
> instance2
```



```
[ubuntu@workstation (keystone-admin)]:~$ openstack server create \
> --image ubuntu \
> --flavor m1.small \
> --network shared \
> --security-group dev-secgroup \
> --key-name dev-keypair \
> --user-data ~/hello \
> instance2
+-------------------------------------+--------------------------------------+
| Field                               | Value                                |
+-------------------------------------+--------------------------------------+
| OS-DCF:diskConfig                   | MANUAL                               |
| OS-EXT-AZ:availability_zone         |                                      |
| OS-EXT-SRV-ATTR:host                | None                                 |
| OS-EXT-SRV-ATTR:hypervisor_hostname | None                                 |
| OS-EXT-SRV-ATTR:instance_name       |                                      |
| OS-EXT-STS:power_state              | NOSTATE                              |
| OS-EXT-STS:task_state               | scheduling                           |
| OS-EXT-STS:vm_state                 | building                             |
| OS-SRV-USG:launched_at              | None                                 |
| OS-SRV-USG:terminated_at            | None                                 |
| accessIPv4                          |                                      |
| accessIPv6                          |                                      |
| addresses                           |                                      |
| adminPass                           | kjQayg4tCXV6                         |
| config_drive                        |                                      |
| created                             | 2025-07-07T16:26:06Z                 |
| flavor                              | m1.small (2)                         |
| hostId                              |                                      |
| id                                  | ccd1fecd-537a-43ad-8a9e-592ec33078e6 |
| image                               | ubuntu (329d361e-f6dc-               |
|                                     | 4b72-b200-3de0ec230e65)              |
| key_name                            | dev-keypair                          |
| name                                | instance2                            |
| progress                            | 0                                    |
| project_id                          | 39e851b14f864573aad60582c35e40dc     |
| properties                          |                                      |
| security_groups                     | name='381cc27c-6434-42d2-b10c-       |
|                                     | 70c228a4c976'                        |
| status                              | BUILD                                |
| updated                             | 2025-07-07T16:26:05Z                 |
| user_id                             | 14f5376f00c04e90b7103dd8d4263040     |
| volumes_attached                    |                                      |
+-------------------------------------+--------------------------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

3.4. Verify that the status of the **instance2** instance is **ACTIVE**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server list
+--------------------------------------+-----------+--------+------------------------+--------+----------+
| ID                                   | Name      | Status | Networks               | Image  | Flavor   |
+--------------------------------------+-----------+--------+------------------------+--------+----------+
| 352cc7e5-8e0a-4bc7-946e-aac5da0bb683 | instance2 | ACTIVE | shared=192.168.233.16  | ubuntu | m1.small |
+--------------------------------------+-----------+--------+------------------------+--------+----------+
[ubuntu@workstation (keystone-admin)]:~$
```

3.5. List the floating IPs to see the free address, which was disassociated from the previous instance when the instance was deleted.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip list
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack floating ip list
+-------------+---------------------+------------------+------+------------------+---------------+
| ID          | Floating IP Address | Fixed IP Address | Port | Floating Network | Project       |
+-------------+---------------------+------------------+------+------------------+---------------+
| 9c8804cd-23 | 172.25.250.76       | None             | None | 8f37949a-cf0a-   | 39e851b14f864 |
| 18-405b-ac4 |                     |                  |      | 4df1-86fd-       | 573aad60582c3 |
| 5-fc89e0b7a |                     |                  |      | 8581a36a573a     | 5e40dc        |
| 509         |                     |                  |      |                  |               |
+-------------+---------------------+------------------+------+------------------+---------------+
[ubuntu@workstation (keystone-admin)]:~$
```

3.6. Assign the floating IP to **instance2**.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server add floating ip \
> instance2 172.25.250.76
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server add floating ip \
> instance2 172.25.250.76
[ubuntu@workstation (keystone-admin)]:~$
```

3.7. Verify that the instance was assigned the floating IP address.

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server list \
> -c Name \
> -c Networks
```

```
[ubuntu@workstation (keystone-admin)]:~$ openstack server list \
> -c Name \
> -c Networks
+-----------+--------------------------------------+
| Name      | Networks                             |
+-----------+--------------------------------------+
| instance2 | shared=192.168.233.16, 172.25.250.76 |
+-----------+--------------------------------------+
[ubuntu@workstation (keystone-admin)]:~$
```

3.8. Use the **scp** command to copy the **~/Downloads/dev-keypair.pem** file to the **devstack** machine. When prompted to enter the password for **ubuntu@192.168.1.20**, enter **ubuntu**.

```
[ubuntu@workstation (keystone-admin)]:~$ scp ~/Downloads/dev-keypair.pem \
> 192.168.1.20:~/dev-keypair.pem
```

```
[ubuntu@workstation (keystone-admin)]:~$ scp ~/Downloads/dev-keypair.pem \
> 192.168.1.20:~/dev-keypair.pem
ubuntu@192.168.1.20's password:
dev-keypair.pem                              100% 1676     2.6MB/s   00:00
[ubuntu@workstation (keystone-admin)]:~$
```

3.9. SSH into the **devstack** machine. Enter **ubuntu** when prompted for a password.

```
[ubuntu@workstation (keystone-admin)]:~$ ssh 192.168.1.20
```

```
[ubuntu@workstation (keystone-admin)]:~$ ssh 192.168.1.20
ubuntu@192.168.1.20's password:
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-94-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings

Last login: Fri Feb  9 22:37:16 2024
ubuntu@devstack:~$
```

3.10. SSH into **instance2** with the **dev-keypair** private key.

```
ubuntu@devstack:~$ ssh -i ~/dev-keypair.pem 172.25.250.76
```

```
ubuntu@devstack:~$ ssh -i ~/dev-keypair.pem 172.25.250.76
The authenticity of host '172.25.250.76 (172.25.250.76)' can't be established.
ED25519 key fingerprint is SHA256:rl4Akqw7dM+Hv79JX5xvY2Nj3pxV9eR4ZIYZRT1ZPww.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.25.250.76' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-92-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

  System information as of Thu Jun 20 14:46:06 UTC 2024

  System load:  0.5302734375      Processes:             85
  Usage of /:   7.4% of 19.20GB   Users logged in:       0
  Memory usage: 8%                IPv4 address for ens3: 192.168.233.16
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@instance2:~$
```

Note

It may take several minutes for the instance to fully boot and be available for an SSH connection. Until then, the connection will be refused.

3.11. Check **/var/log/cloud-init.log** to confirm that the **cloud-init** script ran.

```
ubuntu@instance2:~$ sudo tail /var/log/cloud-init.log
```

```
ubuntu@instance2:~$ sudo tail /var/log/cloud-init.log
2025-07-07 16:33:07,291 - util.py[DEBUG]: Writing to /var/lib/cloud/instance/boot-finished - wb: [644] 70 bytes
2025-07-07 16:33:07,302 - handlers.py[DEBUG]: finish: modules-final/config-final_message: SUCCESS: config-final_message ran successfully
2025-07-07 16:33:07,304 - main.py[DEBUG]: Ran 11 modules with 0 failures
2025-07-07 16:33:07,310 - atomic_helper.py[DEBUG]: Atomically writing to file /var/lib/cloud/data/status.json (via temporary file /var/lib/cloud/data/tmpjqk3qwy6) - w: [644] 592 bytes/chars
2025-07-07 16:33:07,316 - atomic_helper.py[DEBUG]: Atomically writing to file /var/lib/cloud/data/result.json (via temporary file /var/lib/cloud/data/tmptn_pm9_k) - w: [644] 87 bytes/chars
2025-07-07 16:33:07,319 - util.py[DEBUG]: Creating symbolic link from '/run/cloud-init/result.json' => '../../var/lib/cloud/data/result.json'
2025-07-07 16:33:07,321 - util.py[DEBUG]: Reading from /proc/uptime (quiet=False)
2025-07-07 16:33:07,323 - util.py[DEBUG]: Read 13 bytes from /proc/uptime
2025-07-07 16:33:07,325 - util.py[DEBUG]: cloud-init mode 'modules' took 4.088 seconds (4.09)
2025-07-07 16:33:07,326 - handlers.py[DEBUG]: finish: modules-final: SUCCESS: running modules for final
ubuntu@instance2:~$
```

3.12. View the **/etc/hosts** file to ensure that the hostname was appended correctly.

```
ubuntu@instance2:~$ cat /etc/hosts
```

```
ubuntu@instance2:~$ cat /etc/hosts
127.0.0.1 localhost

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
127.0.1.1 instance2
ubuntu@instance2:~$
```

3.13. Ensure that the **/root/hello.txt** file exists and has the correct content.

```
ubuntu@instance2:~$ sudo cat /root/hello.txt
```

```
ubuntu@instance2:~$ sudo cat /root/hello.txt
Hello, world!
ubuntu@instance2:~$
```

3.14. The lab is now complete.